

Executive Technical Recommendation: Optimal pH for Adsorption of 2-Naphthoxyacetic Acid

The adsorption performance of the treatment system was evaluated at pH 3 and pH 9 to determine the optimal operating condition for removing 2-naphthoxyacetic acid from aqueous solutions. The compound is a weak acid with a pKa of 3.55, so solution pH controls its speciation between the protonated, neutral form and the deprotonated, negatively charged form. Applying the Henderson–Hasselbalch relationship, at pH 3 (below the pKa) the molecule sits predominantly in its protonated, neutral state — roughly 78% protonated and 22% deprotonated — whereas at pH 9 (well above the pKa) it is essentially fully deprotonated and anionic, close to 100%. This matters because neutral molecules typically undergo stronger hydrophobic and π – π interactions with the adsorbent surface, while the anionic form is more strongly solvated in water and can be electrostatically repelled, both of which reduce uptake. The analytical wavelength of 325 nm was selected because the UV–Vis spectra at both pH conditions show a clear, well-resolved absorbance peak at that wavelength that remains within a reliable measurement range (absorbance $\approx$ 1.1), making it suitable for tracking concentration through the Beer–Lambert law; it was applied consistently to both conditions because the peak position does not shift with pH.

Adsorption kinetics showed rapid initial uptake followed by a slower, still-rising trend through the final timestamp (460 minutes): the measured adsorption capacity reached 57.54 Mol/g at pH 3 versus 39.96 Mol/g at pH 9, and the adsorption efficiency reached 35% at pH 3 versus 23% at pH 9, so pH 3 outperforms pH 9 by approximately 44% in adsorption capacity and approximately 51% in adsorption efficiency at the last reading. Because neither condition had leveled off, the pseudo-second-order uptake model was fitted to each capacity-time series to estimate the true saturation behavior: at pH 3, the fitted equilibrium capacity is 103.5 Mol/g with a rate constant of 2.58×10^{-5} g/(Mol·min) ($R^2 = 0.995$), and at pH 9 the fitted equilibrium capacity is 106.1 Mol/g with a rate constant of 1.14×10^{-5} g/(Mol·min) ($R^2 = 0.991$). The fitted equilibrium capacities are therefore close to one another — within about 2.5% — unlike the roughly 44% gap seen at the last reading; the meaningful difference is kinetic rather than capacity-based: pH 3's rate constant is more than double that of pH 9, so pH 3 reaches a much larger share of its ultimate capacity within a practical contact time (56% of its fitted equilibrium by 460 minutes, versus 38% at pH 9).

It is therefore recommended that the treatment system be operated at pH 3. While the two conditions may approach a similar ultimate capacity given enough contact time, pH 3 is held largely (about 78%) in its neutral, protonated form, which promotes the stronger surface interactions responsible for its substantially faster uptake and its higher measured capacity and efficiency at any realistic operating timescale; operating at pH 9 drives the molecule almost entirely (close to 100%) to its repelled anionic form, which measurably slows adsorption and lowers performance within the timeframes this system will actually run.